



Introduction to Data Structures and Algorithms



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Outline



1. Resources and Tools
2. What is a “Data Structure”?
3. What is an “Algorithm”?
4. Prerequisites
5. Lesson Plan
6. Grading
7. Suggestions



Resources



Course Materials Website:

https://github.com/noswolf/DSA_BIT/tree/DSA_25



Tools



1. Google Colaboratory (online)
 - Interactive notebooks (.ipynb)
 - <https://colab.research.google.com/>
 - Login with your kmitl email address
2. VSCode or any other IDEs (offline)



What is a “Data Structure” ?



How do we store, organise, and retrieve data on a computer?



What is a “Data Structure” ?



- Way to **store** and **organise** data
- Enable efficient **access** and **modification** of data
- Designed for a specific algorithm
 - Strengths and limitations
 - Time and space complexity



Abstract Data Type



- A data type where only **behavior** is defined but not implementation.
- Examples: Array, List, Map, Queue, Set, and etc.



Common vs Abstract Data Type

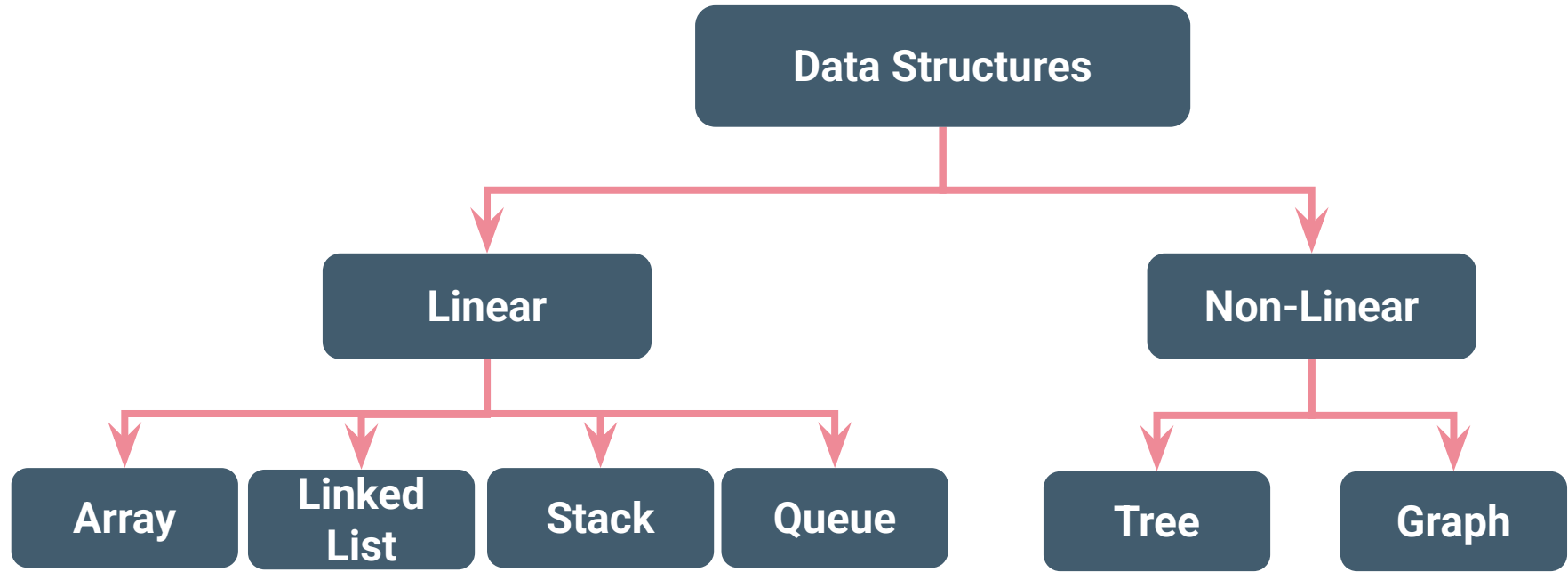
Common

- Integer
- Floating-point number
- Character
- String
- Boolean
- etc.

Abstract

- Array
- List
- Map
- Queue
- etc.

Type of Data Structure



Check out for a comprehensive list of data structures at
https://en.wikipedia.org/wiki/List_of_data_structures



What is an “Algorithm” ?





What is an “Algorithm” ?

- Well-defined procedure or set of instructions to
 - transform input to output or
 - accomplish a task or
 - solve a computational problem





Why care about an “Algorithm” ?

How can we efficiently (in space/time) carry out some typical data processing operations?

How do we analyze and describe their performance?



Example: Sorting numbers





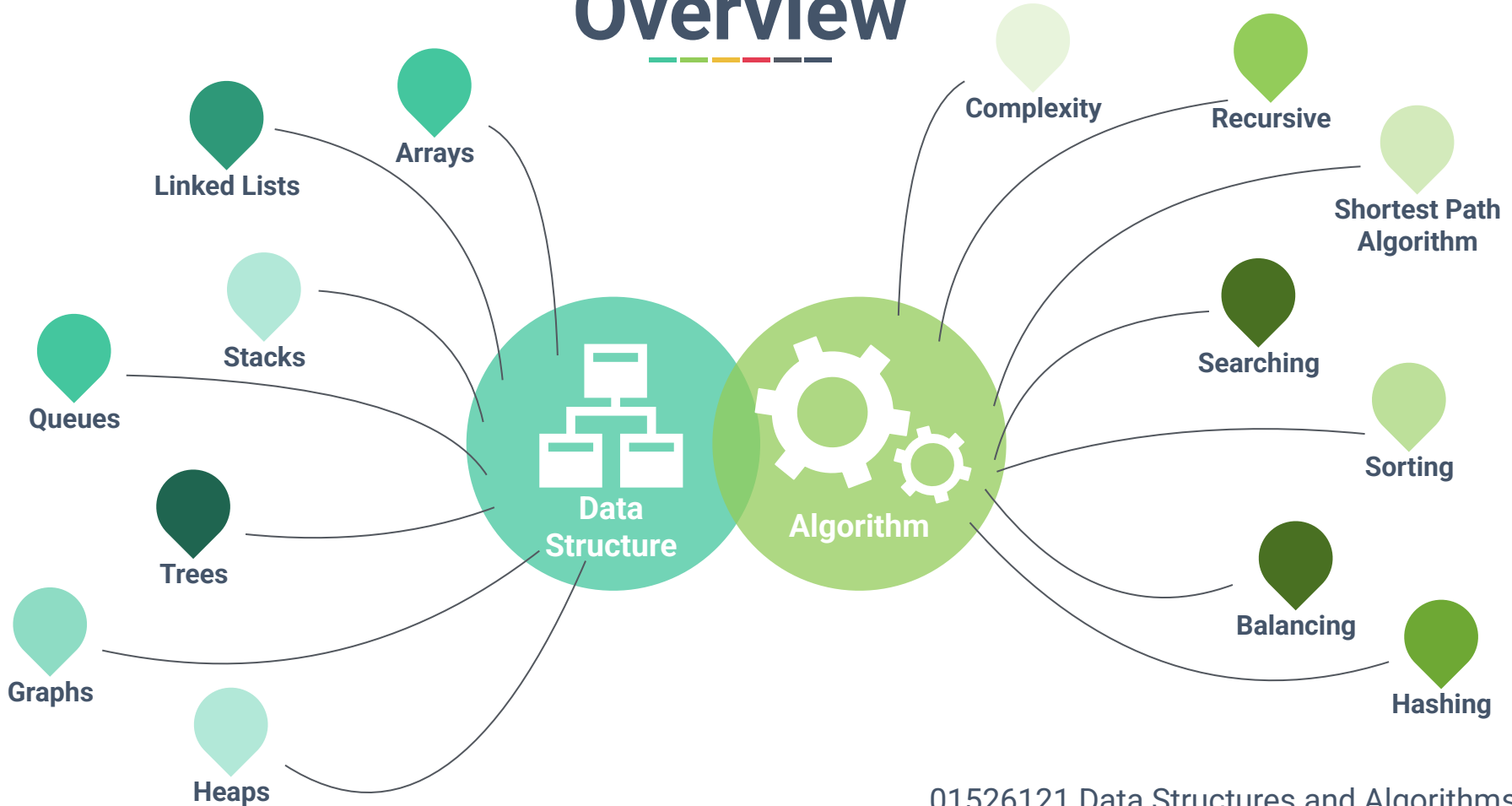
What kind of problems are solved by algorithms?



- Pricing and Valuation Tasks
 - Computing fair prices of derivatives (e.g., options, futures)
- Portfolio Optimization & Asset Allocation
 - Finding the best mix of assets for maximum return at acceptable risk
- Internet: Routing, searches, and security
 - **Shortest path**, search engines, encrypted communication
- E-commerce
 - Ads, recommendations, authentications



Overview





Summary



- Data Structure
 - Way to store and organise data, allowing operations to be performed efficiently.
- Algorithm
 - Step-by-step procedure, which performs on data structure, to be followed to solve a problem/accomplish a task.



Prerequisites

- Fluent in Python Programming
- Comfortable with development processes
 - Writing a function
 - Debugging and testing a code

Lesson Plan (till Midterm)

*Public holidays are highlighted in red

Week	Topics	Labs
24/11/2025	Introduction	#1
01/12/2025	Recursion (Self-revision)	Homework
08/12/2025	Algorithm Analysis	#2
15/12/2025	Arrays & Binary Search	#3
22/12/2025	Sorting	#4
29/12/2025	-	-
05/01/2026	Stacks	#5
12/01/2026	Queues	#6



Lesson Plan (after Midterm)

*Public holidays are highlighted in red

Week	Topics	Labs
26/01/2026	Linked List	#7
02/02/2026	Linked List (Cont.)	
09/02/2026	Trees	#8
16/02/2026	Binary Search Trees	#9
23/02/2026	Graphs	#10
02/03/2026	Graphs (Cont.)	
09/03/2026	-	-
17/03/2026	Final Exam	



Grading

Attendance	5%
Lab Assignment	20%
Midterm Exam	35%
Final Exam	40%

Attendance Score

Out of 13 times, If you attend onsite

12+ times = 5%

11 times = 4%

10 times = 2%

9 times = Unable to take a final exam

Note: Leave notice must be submitted at least 48 hrs before the lecture.

Note 2: Lab submission != Checked Attendance



Lab Assignment

Each lab required to be submitted and presented to TAs during the lab session only

- Late submission is not allowed and will not be graded
- Your TAs will be your grader, not your instructor
- If you need any help, feel free to ask me or your TAs
- MS Teams submission is required for your own benefit.
- You can ask your instructor to see a checklist of your submissions.



Suggestion

Working on Lab tasks

- Although this course is intense on programming, you are encouraged to work on lab with your classmates, including having intense discussion.
- If you are stuck, unsure about the task, or have no idea about the error message, feel free to ask TA and they are happy to help.



Suggestion

Generative AI age

- Even though AI features are allowed and available for you to use freely in this course, please restrain from using it as a first thing when you attempting any task.
- Please Turn off AI generative features in your Colab environment, otherwise you are going to see solutions just as you starting to type.
- It can be very useful if you run out of idea/solution or need a hint from it, including help prepare you for the exam.



Things to Seriously Consider

- **Recording** my lectures in any capacity **is not allowed**.
- However, you are allowed to
 - Take photo of my handwritings on the whiteboard or
- Alternatively, Public lectures on YouTube is a great material for you to understand the complex data structures concepts.



Reading List

Essential

Goodrich, M.T., Tamassia, R. and Goldwasser, M.H., 2013. ***Data structures and algorithms in Python***. John Wiley & Sons Ltd.

Recommended

Cormen, T.H., Leiserson, C.E., Rivest, R.L. and Stein, C., 2022. ***Introduction to algorithms***. MIT press.

Miller, B.N. and Ranum, D.L., 2011. ***Problem solving with algorithms and data structures using python***, 2nd ed. Franklin, Beedle & Associates Inc.